

Smartphone Market Analysis Project Report (2026)

1. Project Overview

What is this project?

This project is a helpful tool made for regular people who want to buy a new smartphone but get confused by all the choices. Nowadays, when you try to buy a phone, companies throw a lot of tricky marketing words, weird tech numbers, and different prices at you. It is really hard for a normal buyer to know if a phone is actually worth the money or if they are just paying extra for a famous brand name.

The main goal of this project is to fix this problem. We wanted to make a clear, honest guide that uses real facts to help shoppers find the perfect phone for their budget and needs. We used a big list of data that has information on over a thousand different phones. This dataset already includes basic details like battery size and cameras, as well as two helpful scores: a "Spec Score" (how powerful the phone is) and a "Value for Money Score" (if it is worth the price).

My job in this project was not to invent those scores, but to take that big list of raw data and turn it into something useful. Here is what I did:

- **1. Cleaned the Data:** I used Excel to fix mistakes, clear out broken information, and make sure everything was neat.
- **2. Dug Deep with SQL:** I wrote SQL code to sort through the data and find hidden details, like matching up phones in each price group to see which ones are truly faster or better than average.
- **3. Built a Dashboard in Power BI:** I created an easy-to-use screen with charts where users can click around, change filters, and compare phones instantly.

In the end, this project works like a smart shopping assistant. Whether someone is a gamer who wants a fast phone, a creator looking for a great camera, or just someone trying to find a cheap but good phone, this project helps them see past the commercial ads and make a smart choice based on real facts.

2. Dataset Overview

What data does it have?

The dataset is basically a big list that holds all the technical details, prices, and scores for many different mobile phones.

- **Total Rows:** 979 distinct smartphone models (after clearing out luxury outliers).

- **Total Columns:** 22 features / detailed specs for each smartphone profile.

3. Data Dictionary

Here is an easy guide to what columns are in the dataset, what kind of data they hold, and what they mean:

Basic Info & Groups

- **brand_name (Text):** The company that made the phone (like Apple, Samsung, or Xiaomi).
- **model (Text):** The specific name of the phone model.
- **price (Number):** How much the phone costs.
- **price_category (Text):** The tier group the phone fits into based on cost (Budget, Mid-Range, Premium, or Flagship).

Performance Scores (Already in the Data)

- **spec_score (Number):** A score that shows how powerful the phone's hardware parts are.
- **vfm_score (Number):** "Value For Money" score. It shows if you get a good amount of features for the price tag.

Inside Parts (Processor & Memory)

- **processor_brand (Text):** The company that made the main processing chip (like Qualcomm, MediaTek, or Apple).
- **processor_name (Text):** The exact retail name of the main chip inside the device.
- **num_core (Number):** How many logical cores the chip uses (more cores mean better multi-tasking).
- **processor_speed (Number):** How fast the chip runs natively, measured in GHz.
- **ram (Number):** The phone's short-term memory, measured in Gigabytes (GB), which helps apps run smoothly.
- **rom / memory (Number):** The internal storage capacity for photos, videos, and apps, measured in Gigabytes (GB).

Battery & Power Delivery

- **battery_capacity (Number):** How big the battery cell is, measured in milliampere-hours (mAh). Larger numbers mean longer stamina.
- **fast_charging (Number):** The maximum supported charging speed provided by the charger, measured in Watts (W).
- **charging_ratio (Number):** A calculated number matching charging speed with battery size to show operational efficiency.
- **charging_speed_type (Text):** A clean label indicating if the phone uses Slow Charging, Standard Charging, or Super Fast Charging.

Multimedia (Display & Camera)

- **screen_size (Number):** The corner-to-corner diagonal measurement of the physical display panel, in inches.
- **refresh_rate (Number):** How many times the screen updates its visual picture per second, measured in Hertz (Hz).
- **num_rear_cameras (Number):** The physical count of dedicated camera lenses mounted on the back side.
- **rear_camera (Number):** The megapixel (MP) size resolution of the primary rear camera lens sensor.
- **front_camera (Number):** The megapixel (MP) size resolution of the front-facing selfie camera lens sensor.

To download your original data:

👉 [Click here to view the Unclean Dataset \(Kaggle Source\) on GitHub](#)

(This is the raw, unedited list of phones containing the typos and missing details before cleaning.)

4. Tools We Used

To clean the data, find insights, and build the final charts, I used three main tools. Each tool had a specific job to do in this project lifecycle:

- **Microsoft Excel:** I used Excel as the very first step to clean up the data. Before moving the information anywhere else, I used Excel to look at raw files, fix empty or missing details, correct spelling errors, and delete duplicate rows so the foundational data was neat.
- **Microsoft SQL Server (SQL):** I used SQL as the core database tool to search through data and perform mathematical operations. I wrote SQL code scripts to find hidden details, calculate exact middle points (medians) for specific price levels, and rank the top phones based on real stats.
- **Microsoft Power BI:** I used Power BI to make the actual interactive dashboard system. I connected it directly to the clean data and turned rows of raw numbers into clean charts. I also built user selection buttons (slicers) and an optimized Dark Mode layout so users can navigate fields smoothly like using an app.

5. Data Cleaning Summary

Real-world data always has mistakes. I used Excel to fix scraping errors, standardize text formats, and fill in missing gaps so that our database math and Power BI charts would be 100% correct.

Names & Brands

Typos (like "samaung"); split brand names ("red" vs "redmagic"); extra

Corrected spelling to "Samsung", merged gaming models under "Redmagic", and

	tech text in model descriptions like "(8GB RAM + 256GB)".	stripped out redundant specification text inside parentheses from all model titles.
Price Outliers	18 luxury or experimental concept smartphones (such as the ₹489,990 Vertu) heavily warped the normal market averages.	Filtered out all 18 phone models priced above ₹150,000 to keep statistical insights strictly focused on regular consumer markets. Brought the market average down to a realistic ₹34,385.
Processors	36 entries missing a processor brand label, blank core counts, inconsistent spacing formats (like "Gen3" vs "Gen 3"), and text typos ("Dimensit").	Utilized an internal matching strategy based on twin specs and conducted live online leak research to resolve missing items down to a single unknown entry (99.9% complete). Uniformed naming layouts.
Battery & Power	Scraper incorrectly grabbed screen size metrics as battery sizes for certain iPhones (e.g., listing a 6.2 mAh battery). 52 missing charging speeds. Overlapping or messy text labels.	Manually imputed correct iPhone battery ratings (4000/4500 mAh). Filled missing wattages using regional brand and pricing category medians. Consolidated messy labels into three strict groups: Slow, Standard, and Super Fast.
Screens & Lenses	Scraper pulled camera megapixel numbers into display fields for 3 specific iPhones (e.g., "48-inch screen"). Gaps found in refresh rates and sensor ratings.	Restored screen size configurations to genuine dimensions (6.1 and 6.7 inches). Assigned 60Hz and 120Hz refresh profiles to respective base and Pro Apple hardware, and filled camera gaps using price group medians.
Operating Systems	Mismatched versions (such as "Android v14.0" vs "v14") and sci-fi placeholder labels grabbed from unreleased items (like "iOS v27").	Stripped decimal endings across versions for solid table joining and grouping. Lowered futuristic prototype OS placeholders to accurate target launch values (iOS v19 and v20).

Key Takeaway

By focusing on data integrity instead of just deleting rows with blank spots, the final dataset was kept at a highly robust 979 smartphones. Every column was successfully converted into clean, whole numbers or standard text groups, making it perfectly ready for SQL querying and Power BI.

To download your final clean data:

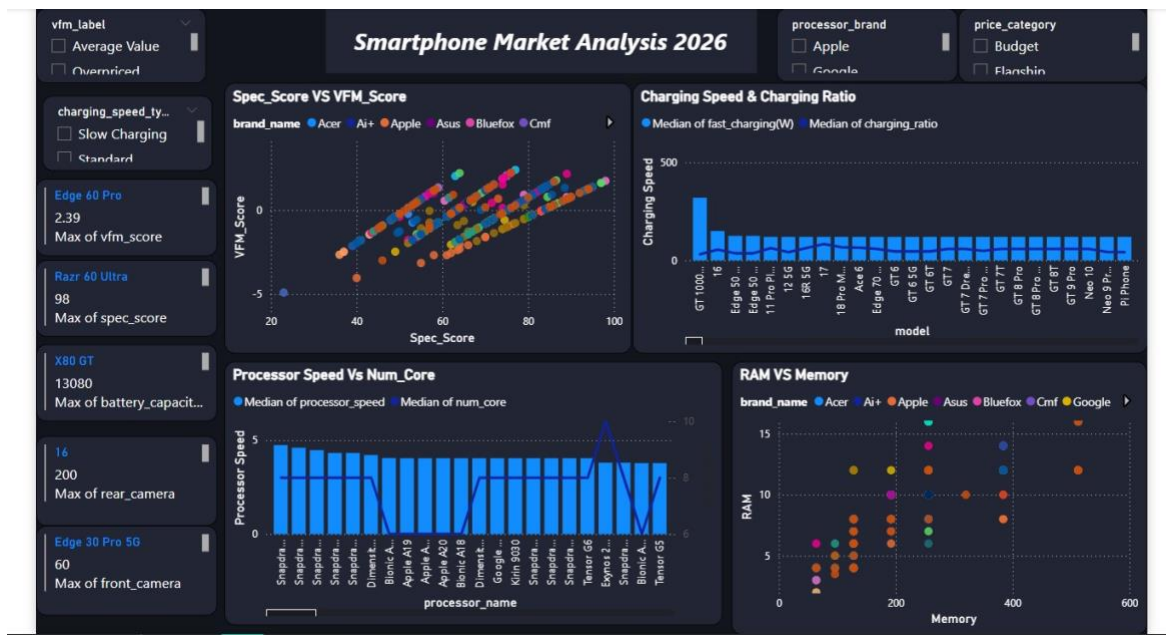
👉 [Click here to view the Cleaned Smartphone Dataset on GitHub](#)

(This is the polished data file after fixing the iPhone battery bugs, charging speeds, and removing extreme luxury price outliers.)

6. Interactive Dashboard Layout & Guide

The Power BI dashboard was designed to look and feel like a modern mobile app. Instead of showing messy tables, it places helpful filters and quick statistics along the top and left, leaving the center open for four easy-to-read, interactive charts.

The Finished Dashboard Screen



To download your interactive dashboard workspace file:

👉 [Click here to download the Power BI Dashboard File \(.pbix\)](#)

(Open this file inside the Microsoft Power BI Desktop application on your laptop to click, filter, and interact with the live phone charts natively.)

How to Navigate and Read the Screen

Everything on this screen is alive and talks to each other instantly. The dashboard is split into three easy-to-use zones:

1. The Control Panels & Leaderboard (Top & Far Left)

- **Top Dropdown Filters:** Use the buttons at the top right (processor_brand, price_category) and top left (vfm_label, charging_speed_type) to instantly focus the entire screen on specific phone groups or budgets.
- **The Quick Leaderboard Cards:** Arranged vertically on the far left, these cards automatically change to show the winning phone names based on your filters. On the main view, you can instantly see the current market leaders: Edge 60 Pro wins for the highest Value for Money (VFM) score; Razr 60 Ultra wins for raw power with a massive Spec Score

of 98; X80 GT crushes the battery capacity rank; and the top cameras hit an incredible 200 MP on the back and a sharp 60 MP for selfies.

2. Power vs. Price Maps (Center-Left Charts)

- **Spec Score VS VFM Score (Top Scatter Plot):** This maps every phone's raw power against its cost-efficiency. Phones sitting in the top-right corner are the ultimate "sweet spot" devices—giving you massive performance without costing a fortune.
- **Processor Speed Vs Num_Core (Bottom Combo Chart):** This chart features solid blue bars tracking the clock speeds (GHz) of different chips (like Snapdragon and Apple Bionic), with a blue line running across them tracking how many CPU cores they use.

3. Power & Storage Clusters (Center-Right Charts)

- **Charging Speed & Ratio (Top Combo Chart):** This chart exposes battery marketing tricks. The bars show charging speed (Watts), while the line running over them shows actual efficiency. Your peak speed champions (like the GT 1000... series) spike high on the far left.
- **RAM VS Memory (Bottom Scatter Plot):** Because phone storage comes in standard blocks (like 8GB or 12GB RAM), the dots form neat vertical groups. It lets you easily pick out the phones sitting at the very top and furthest right, which offer maximum storage space.

 **Interactive User Guide:** If you want to know the exact tech profile for any phone shown in the charts, simply click directly on its dot or bar. The entire dashboard will automatically cross-filter to isolate that single phone's complete specifications across every single chart!

7. Database Analysis Topics & SQL Script Guide

To run a deep analysis on the dataset, I wrote custom code scripts in Microsoft SQL Server. Instead of pulling random numbers, I used custom mathematical logic and filters to search the entire smartphone market for specific trends.

The complete list of investigation topics I successfully built and solved inside the database includes:

Market Groups & Buyer Profiles

- **Price Tier Directories:** Sorting every phone into its exact price group (Budget, Mid-Range, Premium, Flagship).
- **The Deal Finders:** Tracking specific value tiers like Average Value phones, Overpriced devices, Value Kings (high specs for low price), and Performance Beasts (raw power monsters).
- **Feature Checklists:** Instant lists of every phone that has 5G, NFC, or an IR Blaster.

Hardware, Charging & Camera Specs

- **The Core Parts:** Grouping every phone by its exact Processor Name, CPU Speed, RAM, and Internal Storage (ROM).
- **Screen & Battery Power:** Tracking Screen Sizes, Refresh Rates, Charging Speeds, Ratios, and grouping phones into Slow, Standard, or Fast Charging speeds.
- **Camera Systems:** Organizing models by Front Megapixels, Rear Megapixels, and Total Rear Lens Counts.
- **Software:** Grouping all smartphone models by their Operating System (OS) version.

Above-Median Performance Checks (The Smart Filters)

Average numbers can be tricky, so I wrote SQL code to find the exact middle point (median) of the market for each price group. I then filtered for the smart choices—phones that beat that middle line:

- **By Power & Value:** Price category phones that beat the median Spec Score and Value for Money (VFM) Score.
- **By Chip & Memory:** Price category phones that beat the median Processor Speed, Core Count, RAM, and ROM.
- **By Battery & Lens:** Price category phones that beat the median Charging Speed, Charging Ratio, Front Camera, and Rear Camera megapixels.

Technical Code & Result Verification

To keep this report short and clean, the raw data data results and code blocks for these 40+ scenarios are kept in the project repository.

To view your database analysis scripts:

[!\[\]\(2b376d1a92330ab09dad2665d2f89bf5_img.jpg\) Click here to open the Microsoft SQL Server Script on GitHub](#)

(This lets you read the 40+ advanced SQL queries, window functions, and category median math used to find the best phone deals.)

Feel free to open the repository file to see the exact code logic, median calculations, join parameters, and optimizations used to pull the live row answers instantly from the database database engine!